





Integrity Constraints and Update Operations

Ensuring Data Accuracy, Consistency, and Reliability

Learning Outcomes

By the end of our session, we will be able to:

- Define integrity constraints.
- Explain domain, key, entity, and referential integrity.
- Distinguish between primary keys and foreign keys.
- Explain INSERT, UPDATE, and DELETE operations.
- Identify common constraint violations.
- Explain how a DBMS handles referential integrity violations.
- Understand the importance of constraints in real-world databases.



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Why Do We Need Integrity Constraints?

Consider a university database:

RollNo	Name	Department
101	Rahul	CSE
101	Anita	ECE

Problems:

- Can two students have the same Roll Number? ❌
- Can a student belong to a department that does not exist? ❌
- Can a primary key be NULL? ❌

Definition

Integrity constraints are rules that ensure the accuracy, validity, and consistency of data stored in a database.

Objectives

- Maintain data correctness
- Prevent invalid data
- Preserve relationships
- Improve database reliability



Types of Integrity Constraints

Four Fundamental Constraints

- Domain Constraint
- Key Constraint
- Entity Integrity Constraint
- Referential Integrity Constraint

Common SQL Constraints

- NOT NULL
- UNIQUE
- CHECK
- DEFAULT

- Domain: Is this value valid for this attribute?
- Key: Is this record uniquely identifiable?
- Entity integrity: Is the primary key present?
- Referential integrity: Does a foreign key correctly refer to another table?



Domain Constraint

Definition

A domain constraint specifies the set of valid values that an attribute can contain.

Attribute	Valid Domain
Age	18–60
Marks	0–100
Department	CSE, ECE, ME, CE
Gender	Male, Female, Other

Invalid Examples

Age = -5 ❌

Marks = 150 ❌

Department = ABC ❌

Key idea:

Every attribute value must belong to its predefined domain.



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Key Constraint and Entity Integrity

Key Constraint

- A key ensures that tuples can be uniquely identified.

Student_ID	Name
101	Rahul
102	Priya
101	Amit ✗

Entity Integrity

- **Primary Key *must never be NULL.***

Student_ID	Name
101	Rahul
NULL	Priya ✗

Remember the distinction:

Key constraint → uniqueness

Entity integrity → primary key cannot be NULL



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Referential Integrity

Definition

A foreign key must:

Match an existing primary key value in the referenced table, or

Be NULL, if NULL is permitted.

Department

DeptID	Department
10	CSE
20	ECE

Student

RollNo	DeptID
101	10

Invalid

RollNo	DeptID
102	50

Why?

Department 50 does not exist.



Other Common Constraints

Constraint	Purpose	Example
NOT NULL	Value is mandatory	Student Name
UNIQUE	No duplicate values	Email
CHECK	Value must satisfy condition	Age ≥ 18
DEFAULT	Automatic value	Status = Active



Update Operations in the Relational Model

The three basic update operations are:

- **INSERT**

- Adds a new tuple.

Student_ID	Name
101	Rahul
102	Priya

- **UPDATE**

- Modifies existing tuple(s).

INSERT → Add Anita

UPDATE → Change Rahul's email

- **DELETE**

- Removes tuple(s).

DELETE → Remove a student who leaves

Update operation → Constraint checking → Accept or Reject



INSERT Operation and Constraint Checking

INSERT

Adds a new tuple to a relation.

Possible Violations

- Duplicate primary key
- NULL primary key
- Invalid domain value
- Invalid foreign key

DBMS Process

INSERT request → Check constraints → Accept / Reject

The DBMS is responsible for enforcing the integrity rules.

Before

RollNo	Name
101	Rahul

After

RollNo	Name
101	Rahul
102	Anita



UPDATE Operation

UPDATE

Modifies existing tuples.

Possible Violations

- Duplicate key
- Invalid domain value
- Invalid foreign key
- NULL primary key

Example:

Age = -10 ❌

Constraints apply not only when data is inserted, but also when existing data is modified.

Before

RollNo	Email
101	rahul@gmail.com

After

RollNo	Email
101	rahul.cs@gmail.com



DELETE Operation and the Parent-Child Problem

DELETE

Removes tuples from a relation.

Problem

Delete Department 10.



Student 101 still references
DeptID 10.



✗ Referential Integrity Violation

Department-Parent

DeptID	Department
10	CSE

Student-Child

RollNo	DeptID
101	10



Handling Referential Integrity Violations

When a parent record is deleted:

1. RESTRICT

✗ Reject the deletion.

2. CASCADE

🗑 Delete related child records automatically.

3. SET NULL

Set the foreign key to NULL.

4. SET DEFAULT

Replace the foreign key with a predefined default value.

Example

Delete Department 10

→ RESTRICT → Delete rejected

→ CASCADE → Related students deleted

→ SET NULL → Students become unassigned

→ SET DEFAULT → Students receive default department



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Constraint Violations: Operation vs Problem

Operation	Possible Violation
INSERT	Duplicate key
INSERT	Invalid domain
INSERT	Invalid foreign key
UPDATE	Duplicate key
UPDATE	Invalid domain
UPDATE	Invalid foreign key
DELETE	Broken referential integrity

General Process

- User/Application Request → DBMS → Constraint Checking → Accept / Reject



Relational vs NoSQL: Modern Perspective

Relational Databases

Generally provide strong schema and constraint enforcement.

NoSQL Databases

Often provide more flexible schemas.

Integrity rules may be handled through:

Application logic

Schema validation

Database-level features

Key Idea

Relational → stronger built-in integrity enforcement

NoSQL → greater schema flexibility



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Summary

- We learned
- ✓ Domain Constraint → Valid values
- ✓ Key Constraint → Uniqueness
- ✓ Entity Integrity → Primary key cannot be NULL
- ✓ Referential Integrity → Valid table relationships
- ✓ NOT NULL, UNIQUE, CHECK, DEFAULT
- ✓ INSERT, UPDATE, DELETE
- ✓ DBMS checks constraints during modifications
- ✓ Referential actions: RESTRICT, CASCADE, SET NULL, SET DEFAULT



Learning Reflections

1. What are integrity constraints in the Relational Model?
2. Differentiate between key constraint and entity integrity constraint. Why must a primary key be both unique and NOT NULL?
3. What are the INSERT, UPDATE, and DELETE operations in a relational database? Explain the possible integrity constraint violations that may occur during each operation with suitable examples.
4. What is referential integrity? Explain the problem that can occur when a referenced (parent) record is deleted, and describe how RESTRICT, CASCADE, SET NULL, and SET DEFAULT handle such situations.



Homework

- Explain how domain, key, entity integrity, and referential integrity constraints ensure the correctness and consistency of the relations, and describe what happens when INSERT, UPDATE, or DELETE operations violate these constraints, including the roles of RESTRICT, CASCADE, SET NULL, and SET DEFAULT. Use suitable example in your explanation.





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Thanks!

📌 Heads Up!

These slides are just a **guide** — not the full story!

To really learn, make sure to read your book, and explore other resources too.